Hints: Limits – 3. Asymptotics of sequences

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The sequence consists of blocks $1, 2, \dots, m$. Compute the sum $S_n = \sum_{k \leqslant n} a_k$ exactly when $n = T_m = \frac{m(m+1)}{2}$ is the end of the m-th block.

Hint 2. $S_{T_m} = \sum_{j=1}^{m} \frac{j(j+1)}{2} = \frac{m(m+1)(m+2)}{6}$. For $T_m \leqslant n < T_{m+1}$ one has $S_{T_m} \leqslant S_n \leqslant S_{T_{m+1}}$, and $m \sim \sqrt{2n}$.

Hint 3. So $S_n \sim \frac{m^3}{6} \sim \frac{(2n)^{3/2}}{6} = \frac{\sqrt{2}}{3}\, n^{3/2}$. The limit is finite and positive only for $\alpha = \frac{3}{2}$; answer: $(\alpha, \beta) = (\frac{3}{2}, \frac{\sqrt{2}}{3})$.

Problem 2

Note. The condition $0 < x_0 < x_1$ is impossible, since $x_1 = x_0 - x_0^2 < x_0$. The intended condition is $0 < x_0 < 1$; the hints use it.

Hint 1. Show that $0 < x_n < 1$ and x_n decreases, so $x_n \to L$ with $L = L - L^2$, i.e. $L = 0$.

Hint 2. Telescoping (Fact 1 with $p = 2$): $\dfrac{1}{x_{n+1}} - \dfrac{1}{x_n} = \dfrac{1}{1 - x_n} \to 1$. By Stolz–Cesàro (or Cesàro means) $\frac{1}{n x_n} \to 1$, so $\lim n x_n = 1$.

Problem 3

Hint 1. First $x_1 \in (0, 1]$ and x_n decreases to 0. Expand $\dfrac{1}{\sin^2 x} = \dfrac{1}{x^2} + \dfrac{1}{3} + \dfrac{x^2}{15} + O(x^4)$, so $\frac{1}{x_{n+1}^2} - \frac{1}{x_n^2} \to \frac{1}{3}$.

Hint 2. By Stolz, $\frac{1}{n x_n^2} \to \frac{1}{3}$, i.e. $\sqrt{n}\, x_n \to \sqrt{3}$.

Hint 3. For the next term, sum the expansion: $\frac{1}{x_n^2} = \frac{n}{3} + \frac{1}{15} \sum_{k<n} x_k^2 + O(1)$, and $x_k^2 = \frac{3}{k} + o(\frac{1}{k})$ gives $\frac{1}{x_n^2} = \frac{n}{3} + \frac{\ln n}{5} + o(\ln n)$ (in fact $+ C + o(1)$, since the error terms are summable).

Hint 4. Therefore $x_n = \sqrt{\dfrac{3}{n}}\Big(1 - \dfrac{3 \ln n}{10\, n} + O(\frac{1}{n})\Big) = \sqrt{\dfrac{3}{n}} - \dfrac{3\sqrt{3}}{10}\, \dfrac{\ln n}{n^{3/2}} + O(n^{-3/2})$.

Problem 4

Note. As printed the answer is $\lim x_n^3 \ln n = 0$; the natural quantity with a finite nonzero limit is $x_n^2 \ln n \to 3$. The hints derive both.

Hint 1. Put $y_n = x_{n+1}^3 = \frac{1}{n} \sum_{k \leqslant n} \sin^3 x_k$. Then $(n+1) y_{n+1} = n y_n + \sin^3 x_{n+1}$, i.e. $y_{n+1} - y_n = \dfrac{\sin^3 x_{n+1} - x_{n+1}^3}{n+1} < 0$. Deduce that $y_n \downarrow 0$ (if $y_n \to L > 0$, the decrements are $\geqslant c/n$ and sum to ∞).

Hint 2. Since $\sin^3 t - t^3 = -\frac{t^5}{2} + O(t^7)$, we get $y_{n+1} = y_n - \dfrac{y_n^{5/3}}{2(n+1)}(1 + o(1))$. This is a neutral fixed point with an extra factor $\frac{1}{n}$: telescope in the power $y^{-2/3}$.

Hint 3. $y_{n+1}^{-2/3} - y_n^{-2/3} = \dfrac{1+o(1)}{3(n+1)}$, so by Stolz with $b_n = \ln n$: $y_n^{-2/3} \sim \frac{\ln n}{3}$. Hence $x_n^2 \sim \frac{3}{\ln n}$, so $x_n^2 \ln n \to 3$ and $x_n^3 \ln n \sim 3^{3/2}(\ln n)^{-1/2} \to 0$.

Problem 5

Hint 1. No continuity is assumed, so use order instead of the intermediate value theorem: consider $A = \{x \in [a,b] : f(x) \geqslant x\}$, which contains a, and $\xi = \sup A$.

Hint 2. For $x \in A$: $x \leqslant f(x) \leqslant f(\xi)$, so $f(\xi)$ is an upper bound of A and $\xi \leqslant f(\xi)$. Then $f(f(\xi)) \geqslant f(\xi)$, so $f(\xi) \in A$ and $f(\xi) \leqslant \xi$.

Hint 3. Example: $f(x) = 1-x$ on $[0,1]$ has the unique fixed point $\frac{1}{2}$, but $x_0 = 0$ gives $0, 1, 0, 1, \ldots$